



Quality of Tests · Worksheet 1

Further Statistics 1 Chapter 8: Quality of Tests. Answer all questions. Show your working clearly. Total: 45 marks. Time: 55 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are two sections, one for each topic in Chapter 8. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Remember: $P(\text{Type I error})$ is the actual significance level (size) of the test, and $\text{power} = 1 - P(\text{Type II error})$. Give probabilities to 4 decimal places.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Type I and Type II errors	20	
B Power of a test and power functions	20	
Roar-level challenge	5	
Total	45	

Section A Type I and Type II errors

20 marks

1. Explain what is meant by (a) a *Type I error*, (b) a *Type II error*. [2]

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2. A test of $H_0: p = 0.3$ against $H_1: p > 0.3$ uses $X \sim B(20, p)$. The critical region is $X \geq 10$. [2]
Find the probability of a Type I error.

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3. For the test in Question 2, find the probability of a Type II error when the true value of p is 0.5. [3]

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4. A single observation X is taken from $Po(\lambda)$ to test $H_0: \lambda = 5$ against $H_1: \lambda < 5$, using a 5% level of significance. [4]

- (a) Find the critical region and the probability of a Type I error.
- (b) Find the probability of a Type II error when $\lambda = 2$.

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5. The number of faults in a 100 m length of cable has a Poisson distribution with mean λ . A single 100 m length is inspected to test $H_0: \lambda = 6$ against $H_1: \lambda \neq 6$, at the 5% level of significance, with the probability in each tail as close as possible to 2.5%.

(a) Find the critical region for the test.

[4]

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(b) Find the probability of a Type I error.

[1]

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(c) Given that the true value of λ is 8, find the probability of a Type II error.

[3]

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(d) Hence find the power of the test when $\lambda = 8$, and comment on its value.

[1]

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(Total for Question 5 is 9 marks)



Section B Power of a test and power functions

20 marks

6. Explain what is meant by the *power* of a test, and state how it is related to the probability of a Type II error. [2]

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7. A test of $H_0: p = 0.5$ against $H_1: p < 0.5$ uses $X \sim B(10, p)$, with critical region $X \leq 1$. [3]
Find the size of the test and the power of the test when $p = 0.2$.

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8. A single observation X from $Po(\lambda)$ is used to test $H_0: \lambda = 1$ against $H_1: \lambda > 1$. The critical region is $X \geq 3$. [3]
Show that the power function of the test is $1 - e^{-\lambda} \left(1 + \lambda + \frac{1}{2}\lambda^2\right)$, and find the power when $\lambda = 3$.

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9. A test of $H_0: p = 0.4$ against $H_1: p > 0.4$ uses $X \sim B(15, p)$, with critical region $X \geq 10$. [4]

(a) Find the size of the test.

(b) Complete the table of values of the power function.

p	0.5	0.6	0.7	0.8
Power	0.1509			0.9389

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10. The number of emails a manager receives per hour has a Poisson distribution with mean λ . She wants to test $H_0: \lambda = 2$ against $H_1: \lambda > 2$, and considers two tests.

- Test A : count the emails, X , in one hour, and reject H_0 if $X \geq 5$.
- Test B : count the emails, Y , in two hours, and reject H_0 if $Y \geq 8$.

(a) Find the size of each test. [2]

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(b) Find the power of each test when $\lambda = 3$. [4]

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(c) State, giving reasons, which test you would recommend. [2]

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(Total for Question 10 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

11. Two independent observations, X_1 and X_2 , are taken from $Po(\lambda)$ to test $H_0: \lambda = 3$ against $H_1: \lambda < 3$. [5]

H_0 is rejected if $X_1 + X_2 \leq 2$.

(a) Find the size of the test.

(b) Show that the power function of the test is $e^{-2\lambda} (1 + 2\lambda + 2\lambda^2)$, and find the power when $\lambda = 1$.

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END OF WORKSHEET

Total: 45 marks